

software for security flaws. The continuous and broad peer reviews enabled by publicly available source code improve security, because defects that might otherwise be missed are identified and eliminated continuously. Open source software is generally free but, more importantly, the total cost of ownership, including support, is much lower as there is a world of support from the vibrant communities that have evolved around each piece of software."

Agarwalla also believes that open source technology prevents the dependence on vendors and provides flexibility with built-in limitations.

FOSS tools all the way!

Knimbus is built on a J2EE-based platform using technologies like Apache SOLR, MySQL and Perl for internal storage and external connectivity. Other tools used are Struts, Maven, Ibatis, JQuery, etc. In areas like federated search, where Knimbus outperforms any of its competitors, it is empowered by open source technologies like SOLR and Perl.

And you say there are stumbling blocks...

Enterprises that adopt open source technology generally come across a number of bottlenecks in the whole process. But Rahul Agarwalla has a different story to tell. He reveals that, initially, the company was using proprietary solutions, but within a year, the firm migrated Knimbus to Amazon EC2 infrastructure for automatic horizontal and vertical scaling. "We did not face too much trouble in deploying the software tools we were using because we chose stable, reliable and popular OSS. It is important to keep this in mind while selecting OSS because, otherwise, you can run into the following issues:

- Open source software may have multiple licences available. In such cases, the licensing of the software becomes a complex process.
- Lack of proper documentation slows down the learning process for developers, and makes deployment and debugging challenging.
- The death of forums and the abandoning of further development and upgradation can leave you in a fix. Solandra is a typical example of this."

Challenges down the road

Agarwalla is quick to admit that there are quite a few challenges that his team will need to work on in the near future with reference to FOSS. "Quick support in case of a problem could be a real challenge, as we depend on communities and forums. The other issue is the need to upgrade to later versions for bug fixes. This can be complex, as there is no guarantee of update timings or backward compatibility. However, in the

past two years, due to the stability of OSS, we have not faced such issues," says Agarwalla.

And how does Knimbus tackle the issue of support for the deployed OSS? "We are quite close to the Apache community especially in the case of SOLR where many of the committers have helped us. In other cases we use the documentation of the software, community support on forums, and, sometimes, technical support from paid professional experts in the niche areas. We understand that it cannot be a one-way street and thus have given back to the community by way of feedback or code as well as spoken at OS events like Lucene Revolution."

Promoting open source

Open source pundits believe that the basic principle of FOSS is to contribute and give back to the community. Knimbus plays a key role in promoting FOSS at a consumer level and in spreading the spirit of open source in its own ways. "Our technology team has used many open source technologies in the past. When we started Knimbus, selecting open source was a simple choice. As a company, we believe in openness and participation. We are trying to promote the same within the research and development community by providing a cloud-based platform for discovering and sharing open access research, for free," says Agarwalla, with a smile.

And are there any messages to companies that are looking to go the open source route? Agarwalla is quick to respond to this, "Yes, it's very important to keep in mind a few pointers when you plan to consider open source as a viable business model. According to me, these are:

- Before using any open source software, check how active the community is.
- Ensure that it is well documented and supported by the community, for any bugs.
- Look for human resources that have a good knowledge of FOSS, and who are willing to work on FOSS customisation and tweaking.
- Last, but not the least, before deciding on a particular OSS, one should refer to the case studies of other companies already using the solution."

The open source model has a lot to offer to businesses, and it is turning out to be a preferred choice of many enterprises due to its high reliability factor. It will be interesting to see if businesses, irrespective of their size, realise the potential that FOSS provides in the years to come.

So, has the Knimbus case convinced you to take your enterprise on the 'open' road to success? 

By Priyanka Sarkar

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